Quiz 12 Solutions

STAT 109: Introductory Biostatistics

Quiz 12 solutions

Part A: Independence in study design (paired or not paired)

A1. Thirty students have resting heart rate measured before and after a 6-week exercise program. Are the observations paired or independent?

Answer: a (paired/dependent)

A2. Two separate STAT 109 sections take different quiz versions, and each student appears in only one section. Are the scores paired or independent?

Answer: b (independent/not paired)

A3. Each smoker is matched to one nonsmoker of similar age and sex, then lung function is compared. Are measurements paired or independent?

Answer: a (paired/dependent)

A4. Seedlings are randomized to fertilizer A, B, or C, with one fertilizer per seedling. Is the plant height data paired or independent?

Answer: b (independent/not paired)

Part B: Descriptive evidence for ANOVA and chi-square

B1. One-way ANOVA visual evidence: boxplots show visibly different treatment medians, with one group shifted higher. What is the best interpretation?

Answer: a (descriptive evidence of at least one group mean difference)

B2. `anova(fit)` gives p-value = 0.012 at `alpha = 0.05`; what is the best conclusion?

Answer: a (reject `H0`; evidence at least one population mean differs)

B3. Chi-square descriptive evidence: 100% stacked bars show different disease-status proportions across exposure groups. Best interpretation?

Answer: a (descriptive evidence variables may not be independent)

B4. `chisq.test(tab)` gives p-value = 0.18 at `alpha = 0.05`; best conclusion?

Answer: b (fail to reject `H0`; no evidence that at least one pair of groups have a difference in proportions of success)

Part C: Interpreting R code output (conceptual)

C1. For one-way ANOVA with 4 groups, what is the null hypothesis?

Answer: a (`H0`: all four population means are equal)

C2. For chi-square independence between treatment and side-effect category, what is `H0`?

Answer: a (`H0`: all proportions of side-effects are the same between treatment groups)

C3. "If the null were true, data this extreme would be _____" for a small p-value.

Answer: b (uncommon / unlikely)

C4. Best wording for ANOVA with `p = 0.003`.

Answer: a (strong evidence against equal means in the population; the differences in means we see in the data are unlikely by chance alone under `H0`)

Part D: Generalizing to a population (sampling design)

D1. Convenience sample from one gym: can results be generalized to all city adults?

Answer: b (not reliably; convenience sampling limits generalization)

D2. Simple random sample from complete city list: best statement about generalization?

Answer: a (SRS supports population generalization)

Part E: Causation, confounding, and chance (study type)

E1. In an RCT with a very small p-value, what is the best interpretation?

Answer: a (random assignment helps support causal interpretation)

E2. Observational study shows soda intake associated with BMI; best interpretation?

Answer: a (could be causation, confounding, or chance; causation not established by design alone)

E3. Convenience-sample observational study with `p < 0.01`; best interpretation?

Answer: b (Confounding is a concern so we can't assume causation AND generalization to a larger population is not wise because the sample is likely biased)